

WASP-77Ab

R-1452

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-77_170129 · created 2026-08-01T09:39:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457782.65242 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.048
Transit depth (Rp/Rs)^2	2.74 +/- 1.58 [%]
Orbital Inclination (inc)	86.39 +/- 1.44
Airmass coefficient 1 (a1)	0.869 +/- 0.075
Airmass coefficient 2 (a2)	0.064 +/- 0.017
Scatter in the residuals of the lightcurve fit is	9.12 %
Best Comparison Star	#2 - [42, 313]
Optimal Method	PSF photometry
Transit Duration (day)	0.0887 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.165 ± 0.048 vs 0.1301 (deviation 0.73σ)
Duration fit vs published	0.0887 d vs 0.0908 d (deviation 0.37σ)
Mid-time O-C	1.6 min
Depth SNR	3.4
Residual scatter	9.12 %

Light curve

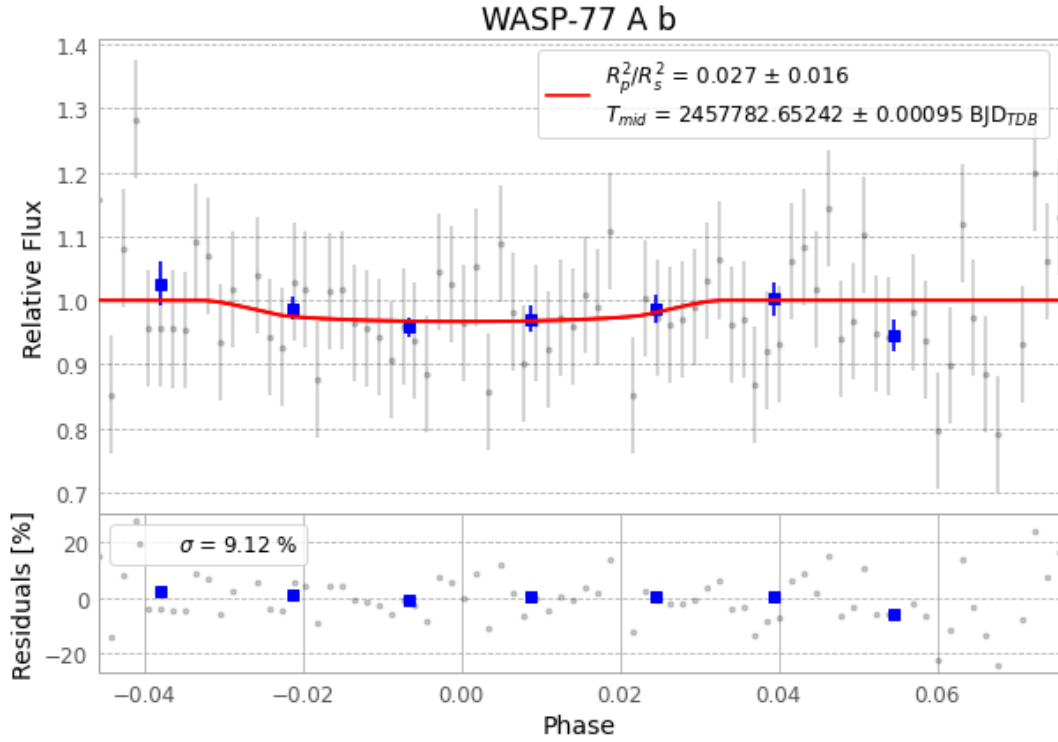


Figure 1 — FinalLightCurve_WASP-77 A b_2017-01-28.png (SHA-256 45aa23fc8cd3852b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 edd9e74ecc6e2e98...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), WASP-77170129020912.FITS → WASP-77170129060612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-170129.log (550fe74adef3...), FinalLightCurve_WASP-77 A b_2017-01-28.png (45aa23fc8cd3...), FinalLightCurve_WASP-77 A b_2017-01-28.csv (edf5e7b9d0a4...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 09:39Z.